

Task Two – Java Assignment

Instructions

Write a Java application. Each question should be implemented inside a **separate class** named:

- Q1, Q2, Q3, ... etc.

Also create a **Main class** that contains the `main` method, which calls all these question classes.

Question #1

Sum of Numbers

Write a program to find and print the sum of all numbers between **1 and 1000** using:

- `for` loop
 - `while` loop
 - `do/while` loop
-

Question #2 (Evens and Odds)

Write a program to print:

- The **even numbers** between 1 and 50 in **one message dialog box**
 - The **odd numbers** between 1 and 50 in **another message dialog box**
-

Question #3 (Divisibility)

Write a program that reads **two integers** and displays all numbers that are:

- **Odd**
- **Divisible by 5**

The program should handle **all cases**, whether:

- `num1 > num2`
 - `num1 < num2`
-

Question #4 (Series)

Write a Java program to calculate and print the value of the following series:

$$1 + 8 + 27 + \dots + N^3$$

Question #5 (Factorial)

Write a program that reads an integer, then calculates and prints its **factorial**.

Hint:

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

Question #6 (Prime Number)

Write a program that reads an integer and indicates whether it is **Prime** or **Not Prime**.

Part Two: Sentinel Controlled Loops

Question #7 (Average)

Write a program that reads an **unspecified number of student marks**.

- The program should calculate and print the **average** of all entered marks
- The program stops reading input when the user enters **any negative value**

Question #8 (Shapes)

Write a program to print the following shape:

```
1
22
333
4444
```

Rules

- **Do NOT use any AI tools**
- Create a **new folder** with a new Java project inside it
- Name the project folder: **TaskTwo**
- Upload your solution into GitHub Repository, then submit the link.